Clement LEE

Address	School of Mathematics, Statistics and Physics Newcastle University, Newcastle upon Tyne, NE1 7RU, UK
Email	clement.lee@newcastle.ac.uk

Academic statistician with expertise in statistical programming and visualisation in R, and modelling of extremes and networks with computational Bayesian statistics.

WORK EXPERIENCE

Lecturer in Statistics 2022-

School of Mathematics, Statistics and Physics, Newcastle University

Research and teaching. Past administrative role: seminar organiser.

Lecturer in Statistics 2019-2022

Department of Mathematics and Statistics, Lancaster University

Research and teaching. Award: ***Tower of Teaching*** for outstanding teaching in the department.

Research Associate (Statistical Modeller) 2016-2019

School of Mathematics and Statistics, Newcastle University

Undertaken statistical research using data generated by mobile and web technologies developed by the Open Lab, with a focus on social network and citation network analysis.

Research Associate 2015

Department of Mathematics and Statistics, Lancaster University

Derived the optimal scaling of one Markov chain Monte Carlo (MCMC) sampler, and implemented efficient MCMC algorithms for stochastic epidemic models based on such optimal scaling.

Actuarial Intern 2010

BOC Group Life Assurance Company Ltd., Hong Kong

Performed statistical analysis to assist pricing and valuation, and helped with checking the rates of new insurance products.

Assistant in Actuarial Science 2009

Department of Statistics and Actuarial Science, the University of Hong Kong

Developed an Excel software of insurance pricing for teaching and consulting.

EDUCATION

PhD in Statistics 2011-2015

Lancaster University

Thesis title: Statistical Models for Financial Time Series and the House Price Index.

MSc in Statistics 2010-2011

Lancaster University

Passed with Distinction. Awards: ***Chancellor's Medal***, and ***Royal Statistical Society prize*** for outstanding performance.

BSc in Actuarial Science 2006-2009
The University of Hong Kong
1st Class Honours. Awards: ***HKU Worldwide Student Exchange Scholarship***, and ***HKU Foundation Entrance Scholarship***.

PUBLICATIONS

A Spliced Preferential Attachment Model for Degree Distributions in Networks 2025
T. Boughen, C. Lee and V. Palacios Ramirez
arXiv preprint arXiv:2506.18726

Reproduction of "(Re)Emerging Disease and Conflict Risk in Africa, 1997–2019" 2026
M. Ashok, D. Doucet, C. Lee, J. Milliman and T. Papastylianou
Institute for Replication, OSF

Conditional Extremes with Graphical Models 2027
A. Farrell, E. F. Eastoe and C. Lee
Computational Statistics & Data Analysis **225**, 108452

The Hierarchical Stochastic Block Model for Replicated Networks 2026
M. Battiston and C. Lee
Statistics and Computing **36** (4), 180

A Bayesian Nonparametric Stochastic Block Model for Directed Acyclic Graphs 2026
C. Lee and M. Battiston
Journal of Computational and Graphical Statistics **35** (2), 619–630

How to Utilise the Limited Supply of Vaccines for Mpox Control in Thailand among High-Risk GBMSM 2026
S. Jirapanakorn, B. Hunsin, A. Endo, C. Lee, S. Jin, B. S. L. Dickens and T. Theparod
Infectious Disease Modelling **11** (3), 1009–1021

Evidencing Preferential Attachment in Dependency Network Evolution 2025
C. Lee
arXiv preprint arXiv:2509.12135

Software Version Control Using Git: A Primer from UKRN 2025
C. Lee et al.
OSF Preprint

Degree Distributions in Networks: Beyond the Power Law 2024
C. Lee, E. Eastoe and A. Farrell
Statistica Neerlandica **78** (4), 702–718

Estimating the Asphaltene Critical Nanoaggregation Concentration Region Using Ultrasonic Measurements and Bayesian Inference 2021
A. Svalova, D. Walshaw, C. Lee, V. Demyanov, N. G. Parker, M. J. Povey and G. D. Abbott

Scientific Reports **11**, 6698

A Hierarchical Model of Non-Homogeneous Poisson Processes for Twitter Retweets 2020

C. Lee and D. J. Wilkinson

Journal of the American Statistical Association **115** (529), 1–15

Performance and Sensitivities of Home Detection on Mobile Phone Data 2020
M. Vanhoof, C. Lee and Z. Smoreda
Big Data Meets Survey Science: A Collection of Innovative Methods

A Review of Stochastic Block Models and Extensions for Graph Clustering 2019
C. Lee and D. J. Wilkinson
Applied Network Science **4** (1), 122

Validity and Reliability of an Online Self-Report 24-h Dietary Recall Method (Intake24): A Doubly Labelled Water Study and Repeated-Measures Analysis 2019
E. Foster, C. Lee, F. Imamura, S. E. Hollidge et al.
Journal of Nutritional Science, 8

Weaving the Topics of CHI: Using Citation Network Analysis to Explore Emerging Trends 2019
C. Lee, A. Garbett, J. Wang, B. Hu and D. Jackson
Extended Abstracts of the 2019 CHI Conference on Human Factors in Computing Systems, 6

A Network Epidemic Model for Online Commissioning Data 2018
C. Lee, A. Garbett and D. J. Wilkinson
Statistics and Computing **28** (4), 891-904

Optimal Scaling of the Independence Sampler: Theory and Practice 2018
C. Lee and P. Neal
Bernoulli **24** (3), 1636-1652

A Social Network Analysis of Articles on Social Network Analysis 2018
C. Lee and D. J. Wilkinson
arXiv preprint arXiv:1810.09781

ThinkActive: Designing for Pseudonymous Activity Tracking in the Classroom 2018
A. Garbett, D. Chatting, G. Wilkinson, C. Lee and A. Kharrufa
Proceedings of the 2018 CHI Conference on Human Factors in Computing Systems, 7

LearningCircle.io: Lessons Learned from Organising Courses with and without a Dedicated Platform 2018
H. Celina, C. Lee, P. Olivier and A. Kharrufa
EdMedia+ Innovate Learning, 76-81

FUNDING SUCCESSES

Knowledge Exchange Placement 2026
Funder: DICE Network+
Project: Predicting item desirability with machine learning to boost digital platform exchanges

Knowledge Exchange Catalyst 2025
Funder: International Centre for Mathematical Sciences
Project: Turbo-charging the gift economy through geospatial optimisation

PHD SUPERVISION

Clive Drowley 2024-
Geospatial Centre for Doctoral Training, Newcastle University
First supervisor; Topic on geospatial analysis of network flows

Thomas Boughen 2023-
School of Mathematics, Statistics and Physics, Newcastle University
First supervisor; Topic on intersections of extreme value theory, networks & Bayesian nonparametrics

Aiden Farrell 2021-2025
Department of Mathematics and Statistics, Lancaster University
Second supervisor; Topic on the extremal behaviour of the nodes when a network grows

Maarten Vanhoof 2017-2018
School of Computing, Newcastle University
Second supervisor; Thesis title: Geographical Veracity of Indicators from Mobile Phone Data

TEACHING

Undergraduate and Postgraduate Teaching 2023-
School of Mathematics, Statistics and Physics, Newcastle University
Taught modules for BSc and MSc students in Mathematics, Statistics & Data Science, including *Statistical Foundations of Data Science*, *Statistical Learning Methodology*, and *Modern Bayesian Inference*. Developed and delivered new module on *Data Visualisation*.

Postgraduate Teaching 2019-22
Department of Mathematics and Statistics, Lancaster University
Taught modules for MSc students in Statistics and in Data Science, including *R programming*, *Statistical Inference*, *Generalised Linear Models*, and *Extreme Value Theory*.

Graduate Teaching Assistant 2011-2015
Department of Mathematics and Statistics, Lancaster University
Marked assignments, held workshop sessions, and provided help and feedback to students. Modules included *Bayesian Inference*, *Likelihood Inference*, *Time Series Analysis*, and *Stochastic Processes*.

PROFESSIONAL ACTIVITIES

Statisticians for Society Scheme 2024-
Royal Statistical Society
Member of review panel for *pro bono* consultancy projects carried out by fellow RSS statisticians for charity organisations.

Local Lead on Open Research 2024-
UK Reproducibility Network (UKRN) & Newcastle University
Led grassroots activities on open research supported by UKRN, at Newcastle University, including the university's inaugural Open Research Conference, training workshops on computational reproducibility, and a journal club on open research.

INVITED TALKS

Community-Powered Circular Economy: The Role of Data 2026-05-14
Exeter Centre for Circular Economy

Evidencing Preferential Attachment in Dependency Network Evolution 2026-03-19
Statistics Seminar, Department of Mathematics, University of York

Breaking Barriers: Enhancing ECR Engagement in Open Science 2025-03-28
ReproducibiliTea Journal Club, University of York

Lessons from My (S)crappy Reproducibility Journey 2024-09-04
Royal Statistical Society 2024 Annual Conference

Citation Network Analysis of a Computer Science Conference 2020-02-28
The Bartlett Centre for Advanced Spatial Analysis, UCL

A Hierarchical Model of Non-Homogeneous Poisson Processes for Twitter Retweets 2019-01-02
Department of Statistics and Actuarial Science, University of Hong Kong

SKILLS

Programming	R, C++ and Python (elementary)
Text editing	EMACS, LaTeX, and (R)Markdown
Miscellaneous	Git and Bash
Languages	English, Cantonese Chinese (native), Mandarin Chinese (fluent)